

ABOUT THE AUTHOR

Alan is a derivatives practitioner and self-study writer. This guide distils the path he wishes had existed when he first sat for an interview at a derivatives desk — one coherent route from the probability axioms to the risk-system Greeks, with the same care given to the math and the institutional context that surrounds it.

He lives in the United States.

QUANT COURSE

Arbitrage Pricing & Derivatives

∂g/∂t + 1/2σ²S²∂²g/∂S² + rS∂g/∂S - rg = 0

the Black-Scholes PDE

INSIDE

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A self-study companion for the graduate derivatives canon

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A fifteen-chapter study guide

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ABOUT THIS BOOK

Sixteen chapters take the reader from a probability refresher to the trading desk. No-arbitrage pricing is built from replication alone, then extended through stochastic calculus, the Black-Scholes PDE, the full Greek atlas, Monte Carlo, Heston stochastic volatility, calibration, short-rate models, caps and swaptions, and a capstone on VaR, expected shortfall, and coherent risk. Every concept lives next to the figure that visualises it. Real-world case studies thread through the chapters: Black Monday 1987, LTCM, Volmageddon 2018, negative WTI, SVB, the UK LDI mini-budget, and the LIBOR-to-SOFR transition. Written for a self-study reader comfortable with undergraduate probability and real analysis who wants a single coherent path from first principles to senior-quant practice.